Lab Practical 5 (Windows)
To develop a MapReduce program to analyze Titanic ship data and find:
(a) Average age of survivors by gender
(b) Number of non-survivors in each passenger class

Run this ONLY ONCE when setting up Hadoop for the first time!

C5mmand:

hdfs namenode –format

Starting Hadoop Services (Every Time)

Always open CMD as Administrator before running these commands!

Step 1: Start HDFS

Command:

start-dfs.cmd

Step 2: Start YARN

Command:

start-yarn.cmd

Step 3: Exit Safe Mode

Command:

hdfs dfsadmin -safemode leave

Step 4: Check Running Services

jps

Practical 5

Step 1: Create Working Directory
Commands:

```
mkdir C:\titanic
cd C:\titanic
```

Step 2: Download Titanic Dataset
Command:

```
curl -o Titanic.csv https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv
```

Step 3: Upload Dataset to HDFS
Commands:

```
hadoop fs -put C:\titanic\Titanic.csv /
hadoop fs -ls /
```

Part A: Average Age of Survivors by Gender

Step 4: Create AvgAgeMapper.java

Command:

```
notepad DeadMaleMapper.java
```

Paste this code:

```
import java.io.IOException;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Mapper;

public class DeadMaleMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
  public void map(LongWritable key, Text value, Context context)
      throws IOException, InterruptedException {

    String line = value.toString();

    if(line.startsWith("PassengerId")) return;

    String[] str = line.split(",(?=(?:[^\"]*\"[^\"]*\")*[^\"]*$)");

    if(str.length > 5 && str[1].equals("0") && str[4].equals("male")) {

      if(!str[5].isEmpty()) {

        try {
          int age = (int)Float.parseFloat(str[5]);
          context.write(new Text("Dead_Male"), new IntWritable(age));
        } catch(Exception e) {}
      }
    }
  }
}
```

Step 5: Create AvgAgeReducer.java
Command:

```
notepad DeadMaleReducer.java
```

Paste this code:

```
import java.io.IOException;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Reducer;

public class DeadMaleReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
  public void reduce(Text key, Iterable<IntWritable> values, Context context)
      throws IOException, InterruptedException {

    int sum = 0, count = 0;

    for(IntWritable val : values) {
      sum += val.get();
      count++;
```

```
    }

    context.write(key, new IntWritable(sum / count));
  }
}
```

Step 6: Create AvgAgeDriver.java
Command:

```
notepad DeadMaleDriver.java
```

Paste this code:

```
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class DeadMaleDriver {
  public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();

    Job job = Job.getInstance(conf, "Dead Male Avg Age");

    job.setJarByClass(DeadMaleDriver.class);
    job.setMapperClass(DeadMaleMapper.class);
    job.setReducerClass(DeadMaleReducer.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
  }
}
```

Part B: Non-Survivor Count per Passenger Class

Step 7: Create SurvivorMapper.java
Command:

```
notepad FemaleClassMapper.java
```

Paste this code:

```
import java.io.IOException;
import org.apache.hadoop.io.*;
```

```
import org.apache.hadoop.mapreduce.Mapper;

public class FemaleClassMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

  public void map(LongWritable key, Text value, Context context)
      throws IOException, InterruptedException {

    String line = value.toString();

    if(line.startsWith("PassengerId")) return;

    String[] str = line.split(",(?=(?:[^\"]*\"[^\"]*\")*[^\"]*$)");

    if(str.length > 4 && str[1].equals("0") && str[4].equals("female")) {
      context.write(new Text("Class_" + str[2]), new IntWritable(1));
    }
  }
}
```

Step 8: Create SurvivorReducer.java
Command:

```
notepad FemaleClassReducer.java
```

Paste this code:

```
import java.io.IOException;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Reducer;

public class FemaleClassReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

  public void reduce(Text key, Iterable<IntWritable> values, Context context)
      throws IOException, InterruptedException {

    int sum = 0;

    for(IntWritable val : values)
      sum += val.get();

    context.write(key, new IntWritable(sum));
  }
}
```

Step 9: Create SurvivorDriver.java
Command:

```
notepad FemaleClassDriver.java
```

Paste this code:

```
import org.apache.hadoop.conf.Configuration;
```

```
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class FemaleClassDriver {
  public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();

    Job job = Job.getInstance(conf, "Dead Female Per Class");

    job.setJarByClass(FemaleClassDriver.class);
    job.setMapperClass(FemaleClassMapper.class);
    job.setReducerClass(FemaleClassReducer.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
  }
}
```

Compile and Run

Step 10: Compile All Java Files
Command:

```
javac -classpath "%HADOOP_HOME%\share\hadoop\common\hadoop-common-
3.3.6.jar;%HADOOP_HOME%\share\hadoop\mapreduce\hadoop-mapreduce-client-core-
3.3.6.jar;%HADOOP_HOME%\share\hadoop\common\lib\commons-cli-1.2.jar" *.java
```

Step 11: Create JAR Files
Commands:

```
jar cf deadmale.jar DeadMale*.class
jar cf femaleclass.jar FemaleClass*.class
```

Step 12: Delete Old Outputs (if exist)
Commands:

```
hadoop fs -rm -r /output1
hadoop fs -rm -r /output2
```

Step 13: Run Average Age Job
Command:

```
hadoop jar deadmale.jar DeadMaleDriver /Titanic.csv /output1
```

Step 14: View Average Age Output
Command:

```
hadoop fs -cat /output1/part-r-00000
```

Step 15: Run Non-Survivor Count Job
Command:

```
hadoop jar femaleclass.jar FemaleClassDriver /Titanic.csv /output2
```

Step 16: View Non-Survivor Count Output
Command:

```
hadoop fs -cat /output2/part-r-00000
```

Stopping Hadoop Services

Always stop Hadoop properly before closing CMD windows!

Step 1: Stop YARN

stop-yarn.cmd

Step 2: Stop HDFS

stop-dfs.cmd